

Tangency portfolios using graph neural networks

Bin Liu^a, Haolong Li^a, Linshuang Kang^{b,*}

^a Center of Statistical Research, School of Statistics and Data Science, Southwestern University of Finance and Economics, Chengdu, 610000, Sichuan, China

^b Electronic Engineering Institute, National University of Defense Technology, Hefei, 230000, Anhui, China

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ABSTRACT

According to modern portfolio theory, the weights of tangency portfolios are solely determined by the expected returns and the covariance matrix of asset returns. However, estimating expected returns and the covariance matrix poses significant challenges, especially when the number of assets is large. Considering the supply-demand relationships between companies issuing stocks, we propose that incorporating industry chain relationships can enhance the accurate estimation of the covariance matrix. Specifically, we present a method that employs Graph Neural Networks (GNNs) to estimate tangency portfolio weights by aggregating stock features based on the industry chain graph and using the aggregated features to estimate the expected returns and the covariance matrix. In addition to incorporating additional industry information, we propose two strategies to enhance the efficiency of estimation: 1) Calculating the dynamic modularity of the stock relationship graph using aggregated node features and constraining the estimated correlations to exhibit a clustered structure by minimizing modularity. 2) Adding a historical ranking regularization to the expected returns. We validate our approach on two daily stock datasets, demonstrating that our method effectively predicts portfolio returns and Sharpe ratios.

1. Introduction

The mean-variance efficient (MVE) portfolio is a concept derived from Modern Portfolio Theory, introduced by Markowitz (1952). The tangency portfolio is a specific point on the efficient frontier of MVE, which represents the optimal portfolio for an investor seeking the highest risk-adjusted return by considering the trade-off between risk and return. The key idea behind the tangency portfolio is to find the optimal allocation of assets that provides the highest expected return for a given level of risk. Mathematically, it is characterized by the portfolio allocation that maximizes the Sharpe ratio. This allocation is derived from the expected returns and covariance matrix of the available assets.

A fundamental theory in financial asset pricing is the equivalence of the MVE portfolio and the stochastic discount factor (SDF). That is, maximizing the squared Sharpe ratio of the MVE portfolio is a duality of the minimum variance of the SDF (Hansen & Jagannathan, 1991). According to the duality, it is easy to derive that the portfolio weights w of a set of n assets, namely the solution of tangency portfolio, only depends on the expected returns μ and covariance matrix Σ of asset returns.

However, both of the expected return and covariance are very difficult to estimate in practices. Especially, accurate estimation of covariance matrices of the asset returns with a large number of assets is a very challenge problem in statistics and machine learning (Bickel & Levina,

2008a; Fan et al., 2008; Ledoit & Wolf, 2004; Shikhaliev et al., 2019). We know the common approach for estimating Σ is the sample covariance matrix $\hat{\Sigma} \in \mathbb{R}^{n \times n}$. When n is large, however, $\hat{\Sigma}$ is not a good estimation of Σ (Bickel & Levina, 2008a). Specifically, for large n , the number of parameters grows quadratically, requiring a comparable number of samples to obtain reliable estimates. In contrast, when n is large than then sample scale, the sample covariance matrix is ill-conditioned or singular, leading to numerical instability in portfolio optimization.

The limitations of accurately estimating Σ have prompted researchers to explore alternative estimators suited for different applications. In order to address the challenges arising from the high-dimension scenario in covariance estimation, researchers have identified the necessity of imposing certain structural properties on the covariance matrix (Bickel & Levina, 2008a,b; Chen et al., 2010; Huang et al., 2006). However, in certain real-world scenarios, the correlation relationships between variables may not only depend on their features but also on additional contextual information. Therefore, merely exploring the principles of accurately estimating large-scale covariance matrices often fails to address practical issues, especially in financial problems such as stock return prediction, as their essence lies in the external manifestation of human socio-economic dynamics. Consequently, the discovery and utilization of more information concerning the interrelationships among samples can lead to a more effective estimation of high-dimensional covariance matrices. This approach aligns with the modern asset pricing

* Corresponding author.

E-mail address: Kang_L_S@outlook.com (L. Kang).

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paradigm of exploiting more factors (Fama & French, 1993, 2015; Feng et al., 2020).

Beyond the numerous stock asset factors already identified (Feng et al., 2020), the network of supply and demand relationships among publicly traded companies provides crucial information for characterizing the relationships between stock returns. Effectively leveraging these industry chain relationships can aid in estimating the high-dimensional covariance matrix of asset returns. Specifically, we aggregate stock features based on the industry chain network, where nodes represent stocks and node attributes are various stock factors constructed from stock interface data. The aggregated stock features not only carry cross-sectional data but also incorporate industry relationship information, which we use to estimate the expected returns and covariance matrix.

In addition to incorporating industry chain information, we propose two regularization terms to constrain the estimated covariance matrix and mean vector. First, we calculate the dynamic graph modularity of the stock relationship using aggregated node features and minimize modularity to maintain a clustered structure. Second, we impose a ranking consistency regularization on the estimated expected returns. Under the objective of maximizing the Sharpe ratio, the proposed model must satisfy these two regularization terms. The contributions of this paper can be summarized as follows: 1) Introducing industry chains as new factors and integrating them using GNNs. 2) Proposing a dynamic graph modularity constraint based on GNNs. 3) Introducing a ranking constraint on the mean of returns.

2. Related works

With the development of financial markets and the explosive growth of financial data, extracting useful information from vast datasets and leveraging this information to make informed investment decisions has become a critical research topic in finance. Fama et al. Ross (1976) proposed the Arbitrage Pricing Theory (APT), which posits that risk-free arbitrage opportunities for stable excess returns exist in the market if it has not yet reached equilibrium, and that multiple factors can explain risky asset returns. Building on this, Fama et al. Fama and French (1993) introduced the three-factor model and later the five-factor model Fama and French (2015), which attribute the risk factors influencing asset volatility to the market risk premium, book-to-market ratio, market capitalization, P/E ratio, and dividend yield. These empirical asset pricing models clearly delineate the relationship between various factors and expected market returns, providing a robust theoretical foundation and methodological guidance for understanding market anomalies and volatility. Additionally, Feng et al. Feng et al. (2020) proposed an innovative multi-factor selection approach that systematically evaluates the contribution of each new factor to asset pricing. Despite the proliferation of multi-factor models aimed at explaining asset pricing, the complexity and variability of financial markets suggest that unexplored risk factors may still exist. Therefore, further investigation into these potential risk factors is necessary to more accurately explain asset prices and predict market movements. Ao et al. Ao et al. (2019) proposed a regression-based Mean-Variance (MV) portfolio model to maximize the Sharpe ratio while controlling risk.

Traditional financial models and methods often face significant challenges when dealing with financial data, as they may fail to capture the complexity and non-linear relationships inherent in the data. In this context, the application of deep learning models in the financial domain has been gaining increasing attention. Wang et al. Wang et al. (2020) proposed a hybrid approach that combines Long Short-Term Memory (LSTM) networks with MV models. This approach predicts asset returns and selects high-potential assets by capturing the long-term dependencies of financial time series data, followed by the portfolio optimization of the selected assets. Additionally, the author in Du (2022) introduced a novel MV model that utilizes stationary portfolios constructed from co-integrated stocks. This model not only employs attention-based LSTM models but also incorporates machine learning models such as Support

Vector Machines (SVM) and Random Forests to predict expected returns. In the realm of stock return prediction, integrating deep learning into MV and ω portfolio optimization models has shown promise in enhancing portfolio performance Ma et al. (2021). Furthermore, an end-to-end deep learning framework for asset pricing was proposed by Wu et al. Wu et al. (2021). This framework, known as Equity2Vec, effectively captures long-term and evolving cross-sectional interactions through a graph-based component. While these methods leverage deep learning models to a certain extent, they have not introduced additional external information. Instead, they focus on recomputing factors from different perspectives and dimensions to provide explanations and predictions for asset prices.

In a comprehensive system like the securities market, the returns of stock assets are often influenced by interactions among various pricing factors. As more factors are widely utilized, some studies have begun to integrate diverse asset information as abstract factors. For instance, Huang et al. Huang et al. (2022) proposed a deep learning framework that integrates three distinct types of market information structures to provide a more comprehensive assessment of asset risk. Similarly, Acero et al. Acero et al. (2024) compared the integration of ESG (Environmental, Social, and Governance) criteria with a modified MV method, demonstrating comparable performance. Moreover, numerous interrelations exist among assets, with the industry chain relationships being particularly prominent. These relationships encompass the supply and demand dynamics between upstream and downstream industries. To capture these intricate supply-demand relationships, we incorporate industry chain information into the investment portfolio using GNNs. This approach allows us to effectively model the interactions and dependencies within the industry chain, thereby enhancing the accuracy and robustness of portfolio optimization.

3. Backgrounds

There exists a duality between the stochastic discount factor (SDF) variance and Sharpe ratios¹. Hansen et al. Hansen and Jagannathan (1991) pointed out that this construction method is based on m_{t+1} being a linear combination of unit payoff, and in order to ensure that SDF does not have an impact on asset value without additional returns, i.e. theoretically, investments with no additional risk should maintain their original value. We start with the minimum-variance SDF in the economy that spans n individual asset excess returns, $\mathbf{r}_t = [r_{1,t}, \dots, r_{n,t}]^T$,

$$m_{t+1} = 1 - \mathbf{w}_t^T (\mathbf{r}_{t+1} - \boldsymbol{\mu}_t), \quad (1)$$

where $\boldsymbol{\mu}_t = \mathbb{E}_t[\mathbf{r}_{t+1}]$ represents the conditional expectation of asset excess returns, $\mathbf{w}_t$ are portfolio weights. By plugging the linear SDF in Eq. (1) in the fundamental pricing relation,

$$\mathbb{E}_t[m_{t+1} \mathbf{r}_{t+1}] = 0, \quad (2)$$

(Eq. (A.5) in Appendix A), then the solution to the SDF loading $\mathbf{w}_t$ takes the form of

$$\mathbf{w}_t = \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}_t, \quad (3)$$

where $\boldsymbol{\Sigma}_t^{-1}$ is the conditional inverse covariance matrix of excess returns, $\boldsymbol{\Sigma}_t = \text{Cov}_t(\mathbf{r}_{t+1})$.

4. Model

4.1. Overall framework

The overall architecture of our proposed model is as shown in Fig. 1. It contains three parts: a GNNs embedding module, a mean and precision matrix fitting layer, and a portfolio weight prediction layer. Since each stock asset corresponds to a unique listed company, we propose to use

¹ See Appendix A for details.

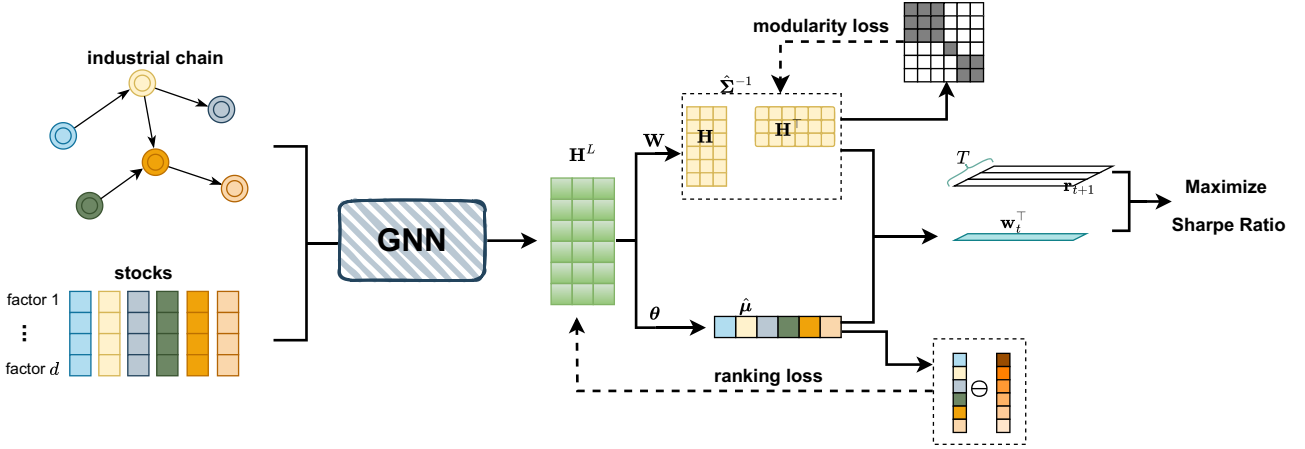


Fig. 1. Model Architecture. The GNN block is designed to aggregate stock factor features over supply-demand networks, ultimately producing a fused embedding for each stock. The proposed method then estimates the inverse covariance $\hat{\Sigma}^{-1}$ and mean $\hat{\mu}$ of returns to predict the portfolio weights. We regularize the estimation process of $\hat{\Sigma}^{-1}$ and $\hat{\mu}$ with graph modularity and ranking consistency, respectively.

the GNNs embedding module with L layers to extract the interdependencies between assets based on the supplying-demanding relationship of the industrial chain, which obtain an embedding matrix $\mathbf{H}^L$. We then fit the expectation μ and inverse matrix of the covariance matrix Σ^{-1} using $\mathbf{H}^L$ as inputs. Finally, we predict the portfolio weights $\mathbf{w}$ based on the modern portfolio theory. We perform *softmax* sorting to approximate the long-short investment strategy. After that, we get the return to maximize the Sharpe ratio. Each module is described in detail as follows.

GNN Embedding Module. Let $\mathbf{A} \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of the supply-demand graph between stock assets, $\mathbf{A}_{ij} = 1$ indicates that company j is a customer of company i , otherwise, $\mathbf{A}_{ij} = 0$. We use $\mathbf{X} \in \mathbb{R}^{n \times d}$ to denote the factor features of n companies (assets). The L -layer GNNs accept $\mathbf{A}$ and $\mathbf{X}$ as inputs to perform message-passing to extract the interdependence among assets to produce an embedding matrix $\mathbf{H}^L$ as follows,

$$\mathbf{H}^l = \text{GNN}(\mathbf{A}, \mathbf{H}^{l-1}; \mathbf{W}^l), \text{ with } \mathbf{H}^0 = \mathbf{X}, l = 1, \dots, L, \quad (4)$$

where $\mathbf{W}^l \in \mathbb{R}^{d^{l-1} \times d^l}$ are parameters of GNNs.

Mean and Precision Matrix Fitting Layer. In practice, estimating expected return μ and inverse covariance matrix Σ^{-1} is challenging when the number of stock n is very large. To enhance the efficiency of estimating μ and Σ^{-1} , we propose two strategies:

1. Exploit more information about the correlations from the supplying-demanding relationship. As we introduced before, $\mathbf{H}^L$ aggregates the interdependence between stocks, as well as the temporal information, which can effectively solve the above problem. Then, μ and Σ^{-1} can be estimated as follows,

$$\hat{\mu} = \mathbf{H}^L \theta, \quad (5)$$

$$\hat{\Sigma}^{-1} = \mathbf{H} \mathbf{H}^T \text{ with } \mathbf{H} = \text{Relu}(\mathbf{H}^L \mathbf{W}), \quad (6)$$

where $\theta \in \mathbb{R}^{d^L}$, $\mathbf{W}$ are parameters.

2. Regularize $\hat{\mu}$ and $\hat{\Sigma}^{-1}$ with structure knowledge. Meanwhile, We impose a regularization term on the expected returns. The details of these regularization techniques will be introduced in the following sections.

Portfolio Weights Prediction Layer. Based on modern portfolio theory, the portfolio weights $\mathbf{w}_t$ can be computed from the mean vector $\hat{\mu}$ and the inverse of covariance matrix $\hat{\Sigma}^{-1}$, i.e. $\mathbf{w}_t = \hat{\Sigma}^{-1} \hat{\mu}_t$. To mimic the long-short

investment strategies, we further filter $\mathbf{w}_t$ with a function $h(\cdot)$ (Feng et al., 2018),

$$h(\mathbf{w}_t) = \text{softmax}(\mathbf{w}_t^+) - \text{softmax}(\mathbf{w}_t^-), \quad (7)$$

where $\mathbf{w}_t^+ := -a_1 e^{-a_2 \mathbf{w}_t}$ and $\mathbf{w}_t^- := -a_1 e^{a_2 \mathbf{w}_t}$, and a_1 and a_2 are two tuning parameters. Then, our portfolio return R_t can be computed as,

$$R_t = \mathbf{w}_t^T \mathbf{r}_{t+1}. \quad (8)$$

4.2. Graph modularity regularization for $\hat{\Sigma}^{-1}$

The modularity of a graph measures the strength of the division of the graph into clusters (also called communities) (Newman, 2006). Clusters are groups of nodes within a network that are more densely connected internally than they are with the rest of the network. For a graph with an adjacency matrix $\mathbf{A}$, the graph modularity is defined as,

$$Q = \frac{1}{2m} \sum_{ij} \left[\mathbf{A}_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (9)$$

where k_i and k_j , are the degrees of nodes i and j , m is the total number of edges in the network, $\delta(c_i, c_j)$ is the indicator function, which is 1 if nodes i and j are in the same community and 0 otherwise. Higher values indicating stronger modular structures. Let $\mathbf{C} \in \{0, 1\}^{n \times k}$ be the cluster assignment matrix and $\mathbf{k}$ be the degree vector of the n nodes, let $\mathbf{K} = \mathbf{A} - \frac{1}{2m} \mathbf{k} \mathbf{k}^T$, then the modularity can be reformulated as (Tsitsulin et al., 2023),

$$Q = \frac{1}{2m} \text{Tr}(\mathbf{C}^T \mathbf{K} \mathbf{C}). \quad (10)$$

The precision matrix $\hat{\Sigma}^{-1}$ represents conditional dependencies in a Gaussian graphical model, where non-zero entries indicate edges between variables. Hence, it is popular to take $\hat{\Sigma}^{-1}$ as an adjacent matrix of an undirected graph (Friedman (2009)). However, unlike a traditional adjacency matrix, the precision matrix can have both positive and negative entries, and its magnitude reflects the strength of conditional dependencies rather than direct connections. The negative entries in $\hat{\Sigma}^{-1}$ may complicate the interpretation of edges in the context of modularity, which is designed for non-negative edge weights. In this work, we choose *Relu*($\cdot$) as activate function to keep the values of $\hat{\Sigma}^{-1}$ (Eq. (6)) nonnegative.

Optimizing modularity of $\hat{\Sigma}^{-1}$ helps in uncovering the underlying structure of the industrial chain, revealing how assets cluster together. In this work, we regularize the estimated inverse covariance matrix $\hat{\Sigma}^{-1}$ using graph modularity. We achieve this by allowing the assignment

matrix $\mathbf{C}$ and the matrix $\mathbf{K}$ to be trainable. Specifically, we define the regularization as follows:

$$\mathbf{C} = \text{softmax}(\mathbf{H}), \quad \text{and} \quad \mathbf{K} = \mathbf{H}\mathbf{H}^\top - \mathbf{k}\mathbf{k}^\top = \hat{\Sigma}^{-1} - \mathbf{k}\mathbf{k}^\top, \quad (11)$$

where $\mathbf{k} = \hat{\Sigma}^{-1} \mathbf{1}$, and $\mathbf{1}$ is a all-one vector.

Then we derive a trainable modularity using GNNs,

$$\mathcal{L}_{\text{modu}} = \frac{1}{2\|\hat{\Sigma}^{-1}\|_0} \text{Tr}(\mathbf{C}^\top (\hat{\Sigma}^{-1} - \mathbf{k}\mathbf{k}^\top) \mathbf{C}), \quad (12)$$

in practice, the L_0 -norm $\|\hat{\Sigma}^{-1}\|_0$ is not convenient for gradient calculation, we hence approximate it with L_1 -norm.

4.3. Ranking regularization for $\hat{\mu}$

Estimating the expected return $\mu \in \mathbb{R}^n$ becomes increasingly challenging as the dimensionality rises. Incorporating the supply-demand relationships using $\mathbf{H}^L$ to estimate μ offers a viable solution, as discussed in Eq. (5). However, we can further enhance the estimation of μ by regularizing it with its historical ranking,

$$\mathcal{L}_{\text{ranking}} = \frac{1}{n} \left| \{(i, j) \mid y_i < y_j, \hat{\mu}_i < \hat{\mu}_j, i \neq j\} \right|, \quad (13)$$

where $\mathbf{y} \in \mathbb{Z}^n$ is the ranking of returns n assets over the recent historical period, and $\hat{\mu} \in \mathbb{R}^n$ is the estimation of μ .

The operation of counting pairs in a set is inherently discrete. The indicator function $\mathbb{I}(y_i < y_j, \hat{\mu}_i < \hat{\mu}_j)$ evaluates to 0 or 1, resulting in a step function that has zero gradients almost everywhere. The loss does not provide a smooth gradient with respect to $\hat{\mu}$, which is necessary for backpropagation. Small changes in $\hat{\mu}_i$ or $\hat{\mu}_j$ may flip the condition $\hat{\mu}_i < \hat{\mu}_j$, causing abrupt changes in the loss value without intermediate gradient information.

To address the non-differentiability, we just simply replace the discrete counting-based loss with a differentiable surrogate–Hinge loss that encourages the predicted returns $\hat{\mu}$ to align with the historical ranking $\mathbf{y}$,

$$\mathcal{L}_{\text{ranking}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max(0, \hat{\mu}_j - \hat{\mu}_i + \tau), \quad (14)$$

where $\mathcal{P} = \{(i, j) \mid y_i < y_j, i \neq j\}$ is the set of pairs where asset i is ranked lower than asset j in the historical ranking. $\tau > 0$ is a margin hyperparameter that enforces a minimum difference between $\hat{\mu}_i$ and $\hat{\mu}_j$. The term $\hat{\mu}_j - \hat{\mu}_i$ should be negative if $\hat{\mu}_i > \hat{\mu}_j$, aligning with $y_i < y_j$.

4.4. Overall loss function

Our objective is to estimate the portfolio weights $\mathbf{w}_t$ to maximize the Sharpe ratio. Given the portfolio weights at time t , $\mathbf{w}_t$, the corresponding portfolio return at time t , $R_t = \mathbf{w}_t^\top \mathbf{r}_{t+1}$ can be computed. Thus, for a prediction time window of length T , we obtain a set of T portfolio returns $R := \{R_{t+1}, \dots, R_{t+T}\}$. Based on R , we can calculate the expected return and standard deviation for the window T as follows,

$$\mathbb{E}_t[R] \approx \frac{1}{T} \sum_{\tau=1}^T R_{t+\tau}, \quad \sigma_t(R) \approx \sqrt{\frac{1}{T} \sum_{\tau=1}^T (R_{t+\tau} - \bar{R}_t)^2}, \quad (15)$$

where $\bar{R}_t = \frac{1}{T} \sum_{\tau=1}^T R_{t+\tau}$. Then the Sharpe ratio (SR) can be calculated $SR = \frac{\mathbb{E}_t[R] - R_f}{\sigma_t(R)}$, where R_f is the risk-free return. We aim to maximize the SR while satisfying the conditions discussed in the previous sections. Therefore, the objective function of the proposed model is:

$$\mathcal{L} = \exp(-SR) + \alpha_1 \mathcal{L}_{\text{modu}} + \alpha_2 \mathcal{L}_{\text{ranking}}. \quad (16)$$

Note that the first term, $\exp(-SR)$, is designed to maximize the SR, in contrast to $\exp(-SR^2)$ as utilized in (Feng et al., 2023). The second term $\mathcal{L}_{\text{modu}}$ and third term $\mathcal{L}_{\text{ranking}}$ correspond to the graph modularity and ranking regularizations, respectively, as previously introduced.

5. Synthetic experiments

Due to the highly complex dynamics behind financial data, in this section we assume that stock returns are related to the upstream and downstream relationships of the industrial chains in which enterprises are located. Under this assumption, we utilize a factor model to generate simulated data.

5.1. Data generation

Adjacency Matrix Definition. Let there are c industries, described by an adjacency matrix $\mathbf{A} \in \mathbb{R}^{c \times c}$, where A_{ij} denotes the supply-demand intensity from industry i to industry j . For directed relationships: $A_{ij} > 0$ if industry i supplies to industry j . Edge probability is higher within clusters ($p_{\text{in}} = 0.3$) than between clusters ($p_{\text{out}} = 0.05$).

The adjacency matrix is converted to a correlation matrix $\mathbf{B}$ using a monotonic function, e.g.,:

$$B_{ij} = \begin{cases} 1, & i = j, \\ \frac{\exp(A_{ij}) - 1}{\exp(1) - 1}, & i \neq j. \end{cases} \quad (17)$$

This maps A_{ij} to $[0, 1]$ while preserving supply-demand intensity. To ensure positive definiteness, we may add a regularization term:

$$\mathbf{B}^\top = \mathbf{B} + \lambda \mathbf{I}, \quad \lambda \approx 10^{-6}. \quad (18)$$

Asset-industry mapping and covariance construction. We define an industry assignment matrix $\mathbf{C} \in \mathbb{R}^{n \times c}$, where $C_{ik} = 1$ if asset i belongs to industry k , $i = 1, \dots, n$ and $k = 1, \dots, c$. The return of asset i is modeled as:

$$r_i = \mu_i + \sum_{k=1}^c C_{ik} (f_k + \epsilon_{ik}), \quad (19)$$

where μ_i is the expected return of asset i , $f_k \sim \mathcal{N}(0, \sigma_k^2)$ is the common factor for industry k , $\epsilon_{ik} \sim \mathcal{N}(0, \sigma_{ik}^2)$ is idiosyncratic risk, independent across assets.

The covariance between assets i and j is:

$$\text{Cov}(r_i, r_j) = \sum_{k=1}^K \sum_{l=1}^K C_{ik} C_{jl} \text{Cov}(f_k, f_l) + \delta_{ij} \sigma_{ii}^2, \quad (20)$$

where δ_{ij} is the Kronecker delta. In matrix form, let Σ be the asset covariance matrix, $\mathbf{F}$ the factor covariance matrix ($F_{kl} = \text{Cov}(f_k, f_l)$), then:

$$\Sigma = \mathbf{C}\mathbf{F}\mathbf{C}^\top + \text{diag}(\sigma_1^2, \dots, \sigma_n^2), \quad (21)$$

with $\mathbf{F} = \text{diag}(\sigma_f) \mathbf{B} \text{diag}(\sigma_f)$, and $\sigma_f = [\sigma_1, \dots, \sigma_c]^\top$.

Return generation process. With the preparation work above, we generate the assets' return with factor model as the following steps:

1. **Factor parameters:** Factor means $\mu_f \in \mathbb{R}^c$ (e.g., industry-specific outlooks), and Factor standard deviations: $\sigma_k \sim U(0.1, 0.2)$, Correlation matrix: $\mathbf{B}$ as defined above.
2. **Cholesky decomposition:** Generate factor sequences $\mathbf{f}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{F})$ via:

$$\mathbf{f}^{(t)} = \mathbf{L}\mathbf{z}^{(t)}, \quad \mathbf{z}^{(t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (22)$$

where $\mathbf{L}$ is the lower triangular Cholesky factor of $\mathbf{F}$.

3. **Asset return calculation:** For asset i in industry k at time t :

$$r_i^{(t)} = \mu_i + C_{ik} f_k^{(t)} + \epsilon_i^{(t)}, \quad \epsilon_i^{(t)} \sim \mathcal{N}(0, \sigma_{ik}^2), \quad (23)$$

with expected returns linked to industry supply-demand activity:

$$\mu_i = \mu_k + \theta \cdot \text{rowsum}(\mathbf{A}_k), \quad (24)$$

where $\text{rowsum}(\mathbf{A}_k)$ is the sum of row k in $\mathbf{A}$, and θ is a sensitivity parameter.

Table 1
Synthetic experiment results for GNN-based tangency portfolio.

Scenario	Sharpe ratio	Returns	Volatility
<i>Noise levels</i> ($p_{in} = 0.3, p_{out} = 0.05$)			
$\sigma_{ik} = 0.01$	2.7312	4.5e-3	0.0073
$\sigma_{ik} = 0.05$	1.2332	4.2e-3	0.0125
$\sigma_{ik} = 0.1$	1.1232	3.6e-4	0.0456
<i>Graph Densities</i> ($\sigma_{ik} = 0.05$)			
$p_{in} = 0.2, p_{out} = 0.03$	0.1332	4.5e-4	0.0498
$p_{in} = 0.3, p_{out} = 0.05$	1.2332	4.2e-3	0.0125
$p_{in} = 0.4, p_{out} = 0.07$	1.4391	5.4e-3	0.0035

5.2. Expected results

The results of the synthetic experiments for the GNN-based tangency portfolio reveal distinct patterns under varying noise levels and graph densities. As shown in Table 1, for noise sensitivity analysis with fixed graph density ($p_{in} = 0.3, p_{out} = 0.05$), the Sharpe ratio decreases significantly from 2.7312 to 1.1232 as the noise parameter σ_{ik} increases from 0.01 to 0.1. Concurrently, returns drop from 4.5×10^{-3} to 3.6×10^{-4} , while volatility surges from 0.0073 to 0.0456, indicating that higher noise degrades both risk-adjusted returns and stability.

Regarding graph density effects at a fixed noise level ($\sigma_{ik} = 0.05$), the Sharpe ratio improves from 0.1332 to 1.4391 as p_{in} increases from 0.2 to 0.4 (with p_{out} scaling from 0.03 to 0.07). Returns follow a similar upward trend, rising from 4.5×10^{-4} to 5.4×10^{-3} , while volatility decreases notably from 0.0498 to 0.0035. This suggests that denser graph structures, which better capture inter-industry dependencies, enhance the portfolio's ability to balance returns and risk. The results highlight the critical role of both noise resilience and graph connectivity in optimizing GNN-based portfolio strategies.

6. Experimental results on real-world data

6.1. Datasets and settings

Industry chain data. This paper uses ChinaScope industry chain data, which includes supplier-customer relationship of 1164 industry nodes (Liu et al., 2023). Typically, each industrial node comprises several listed companies. We determine the supply-demand relationships between companies through the industrial chain network. For instance, if industrial node 1 is upstream of node 2, and node 1 includes company A while node 2 includes company B, then company A has a supply relationship with company B. To model this relationship, we establish connections among companies with supply-demand relationships, thereby constructing the adjacency matrix. This matrix effectively captures the supply-demand dynamics within the industry chain.

Stock factor data. There are two stock datasets used in the experiments as shown in Table 2, CSI 300 and A-share. CSI 300 contains daily volume and price of 246 stocks spanning from January 2001 to December 2023, totally 5564 days. We slice the data with a window size of 50, moving one day at a time, resulting in a total of 5515 samples. The first 40 days of each sample is for training, leaving the remaining 10 days for validation. It includes 65 factors. Hence, the shape of the samples are (5515, 40, 246, 65) for training and (5515, 10, 246, 1) for validation.

The A-share is a collection of daily record for 2912 stocks from January 2016 to August 2023 and it contains 161 factors. Each stock has a time span of 1864 days in this dataset. We apply same data slicing strategy for A-share data, resulting in a total of 1815 samples. The shape of the samples are (1815, 40, 2912, 161) and (1815, 10, 2912, 1) respectively as shown in Table 2.

Factor construction. The 65/161 factors of CSI 300/A-share are derived from several price-volume anomaly factors identified in previous litera-

ture, as well as fundamental factors related to financial indicators. For instance, the factor “Vold” is calculated based on the standard deviation of daily trading volume over the past 1, 6, and 12 months (Chordia et al., 2001). “Idskew” is determined by regressing the daily returns over the past 1, 6, and 12 months against the market capitalization-weighted daily returns of the market portfolio (Ang et al., 2006). The skewness of the residuals from this regression represents the idiosyncratic skewness. The details for other factors refer to Appendix B.

Settings. We use the Adam optimizer and set the learning rate of $5e-3$, $\alpha_1 = 1e-3$ and $\alpha_2 = 1e-1$. We train our model on a Ubuntu Server with a Tesla V100 GPU, running in an environment with PyTorch 2.2.1.

6.2. Overall performance

We select the following three methods as baselines:

1. **Fama-3** Fama and French (1993), a linear models include the Fama three-factor model.
2. **OLS** (Ordinary Least Squares), a linear multifactor model.
3. **PCR** Campbell et al. (1998), a principal component regression multifactor model.
4. **OAS** Chen et al. (2010), a shrinkage method of high-dimensional covariance matrix estimation.
5. **LSTM** Hochreiter and Schmidhuber (1997) aggregates the temporal information of factors data and uses linear layers to reduce the feature dimension, ultimately obtaining an estimate of the portfolio weight w .
6. **MAXSER** Ao et al. (2019), an unconstrained regression representation of the mean-variance portfolio.
7. **FDFM** Feng et al. (2018), a fused deep factor model that generates risk factors from firm characteristics to fit cross-sectional returns. And it has an economic-guided objective function that minimizes aggregated realized pricing errors.
8. **DFGN** Feng et al. (2023), it directly estimate the tangency portfolio weights on high-dimensional individual assets. The deep tangency portfolio combines the market factor and a deep long-short factor constructed using a large number of firm characteristics. At present, there are many studies using deep learning models for portfolio selection, but there is relatively little research focus on exploiting and exploring deep factors.

As shown in Table 3, our methods significantly outperform the baselines on A-share dataset (It comprises 2912 assets and 161 factors). Specifically, A-DGNs achieving the highest SR of 2.7796-more than double the best baseline (OAS at 1.1934). GCN follows closely at 1.5604, also surpassing all baselines. Other variants like SSGC (0.8128), Chebynet (0.7016), and GraphSAGE (0.7048) are competitive but lag behind our top performers, still exceeding weaker baselines such as FDFM (0.1349) and DFGN (-1.0652). This highlights the effectiveness of adaptive and graph-convolutional architectures in capturing complex dependencies for better risk-adjusted returns. Our approaches dominate return as well, with A-DGNs leading at $8.6e-3$, followed by GCN ($6.6e-3$) and GraphSAGE ($4.4e-3$). These are substantially higher than the baselines, where OAS ($3.4e-3$) is the strongest, and several baselines show negative or near-zero returns (e.g., PCR at $-1.7e-2$, FDFM at $-3.7e-3$, DFGN at $-3.1e-3$). SSGC ($4.1e-3$) and Chebynet ($6e-4$) are positive but lower among our methods, yet still outperform most baselines. Baselines generally exhibit lower volatility, with MAXSER at a minimal 0.0007 and OLS at 0.0078. In contrast, our methods show higher risk levels, such as GraphSAGE (0.0774) and SSGC (0.0760), which may indicate greater exposure to market fluctuations. However, A-DGNs (0.0618) and GCN (0.0472) balance this with their superior SR, suggesting efficient risk management despite elevated volatility compared to baselines like LSTM (0.0018). Compared with traditional statistical models and LSTM, the shrinkage estimator OAS achieves better results for A-share market data with a large number of assets.

Table 2
Stock datasets.

Dataset	#Stocks	Training sample shape	Validation sample Shape	Test Split
CSI 300 (2001/1–2023/12)	246	(5515,40,246,65)	(5515,10,246,1)	2021/8–2023/12
A-share (2016/1–2023/8)	2912	(1815,40,2912,161)	(1815,10,2912,1)	2022/11–2023/8

Table 3

The overall performance of both the baselines and the proposed model is assessed across three key metrics: sharp ratio (SR), returns, and standard error (risk).

	A-share			CSI 300		
	Sharp Ratio	Returns	Volatility	Sharp Ratio	Returns	Volatility
Fama-3 Fama and French (1993)	0.8643	9.1e-5	0.013	0.1123	2.2e-4	0.0845
OLS	0.9334	1.4e-4	0.0078	0.1567	6.6e-4	0.0167
PCR Campbell et al. (1998)	0.8523	−1.7e-2	0.045	0.1539	5.1e-4	0.0981
OAS Chen et al. (2010)	1.1934	3.4e-3	0.0142	0.5634	9.1e-4	0.0331
LSTM	0.9302	1e-4	0.0018	0.1939	7.1e-4	0.0742
MAXSER Ao et al. (2019)	0.9139	9e-5	0.0007	0.1921	6.8e-4	0.0157
FDFM Feng et al. (2018)	0.1349	−3.7e-3	0.1303	0.2968	−6e-4	0.1016
DFGN Feng et al. (2023)	−1.0652	−3.1e-3	0.0368	−1.5841	−6.4e-3	0.0744
Ours(GCN)	1.5604	6.6e-3	0.0472	0.8569	3.2e-3	0.2968
Ours(A-DGNs)	2.7796	8.6e-3	0.0618	1.1017	4.8e-3	0.0783
Ours(SSGC)	0.8128	4.1e-3	0.0760	0.4318	1.3e-3	0.0558
Ours(Chebynet)	0.7016	6e-4	0.0459	0.6207	3e-4	0.0133
Ours(GraphSAGE)	0.7048	4.4e-3	0.0774	0.4699	4.8e-3	0.1043

For CSI 300, which includes 246 assets and 65 factors, our A-DGNs and GCN again excel, providing superior SR and returns against baselines, with Chebynet offering a low-risk alternative among our variants. As shown in Table 3, A-DGNs again achieve the best SR, 1.1017, followed by GCN (0.8569)-both exceeding the best baseline (OAS at 0.5634) by a significant margin. Chebynet (0.6207) also surpasses OAS, while GraphSAGE (0.4699) and SSGC (0.4318) are comparable to mid-tier baselines like FDFM (0.2968) but still ahead of poorer performers such as DFGN (−1.5841). This reinforces the advantage of our GNN designs in handling large-scale index data for improved risk adjustment. Our methods achieve the highest returns, led by A-DGNs and GraphSAGE (both at 4.8e-3), then GCN (3.2e-3). These outpace baselines like OAS (9.1e-4) and LSTM (7.1e-4), with several baselines showing negative values (e.g., FDFM at −6e-4, DFGN at −6.4e-3). SSGC (1.3e-3) and Chebynet (3e-4) are positive and competitive, demonstrating consistent gains over traditional models. The volatility of baselines vary widely, with OLS (0.0167) and MAXSER (0.0157) showing low risk, while others like FDFM (0.1016) are higher. Our methods exhibit mixed results: Chebynet has the lowest volatility among ours at 0.0133 (better than most baselines), but GraphSAGE (0.1043) and GCN (0.2968) are notably higher, potentially reflecting aggressive strategies. A-DGNs (0.0783) strikes a reasonable balance, contributing to its top SR.

The results in Table 3 indicate that our model performs particularly well on the A-share dataset. A potential reason for this superior performance is that the A-share dataset encompasses 2912 assets and 161 factors, while the CSI 300 dataset includes only 246 assets and 65 factors. Consequently, the A-share dataset covers a more comprehensive range of industrial relationships than the CSI 300 dataset, providing a richer context for the model to leverage the industry chain information effectively. This difference in dataset composition highlights the importance of a more extensive industrial relationship network, which our model is designed to exploit, thereby delivering better performance in predicting portfolio metrics such as SR, returns, and volatility.

A key factor contributing to the superior performance on the A-share dataset is its broader scope, encompassing a larger number of assets and factors compared to the CSI 300 dataset. The A-share dataset's extensive industrial relationship network-spanning 2912 assets-provides a richer context for our model to exploit, enabling more effective aggregation of supply-demand information via the GNN framework. In contrast, the CSI 300 dataset, with only 246 assets, offers a less comprehensive network,

which may limit the model's ability to fully capture complex interdependencies, resulting in relatively lower SRs (e.g., 1.1017 vs. 2.7796 for A-DGNs).

This disparity in dataset composition highlights the critical role of an extensive industrial relationship network in driving the model's predictive power. This is also where our method differs from traditional high-dimensional portfolio methods. Traditional methods either attempt to find a balance between variance and bias in covariance matrix estimation, such as OAS Chen et al. (2010), or theoretically add risk control constraints, such as MAXSER Ao et al. (2019). In contrast, our methods leverage additional information, the industrial network, combined with tailored regularization strategies, allows for more accurate estimation of the covariance matrix $\hat{\Sigma}$ and expected returns $\hat{\mu}$, leading to enhanced portfolio metrics such as higher SRs, improved returns, and controlled volatility. These findings validate the hypothesis that incorporating industry chain information is particularly advantageous in high-dimensional settings, where traditional methods like Fama-3, OLS, and PCR struggle due to estimation errors and limited structural priors.

Impacts of different message-passing methods. We also observed significant differences in Sharpe ratio (SR) performance across different GNN architectures, e.g., A-DGNs at 2.78 vs. Chebynet at 0.70 on the A-share market. As we know, GNNs differ in how they propagate information across nodes. For example, GCN uses a fixed spectral convolution based on the graph Laplacian, while GraphSAGE employs neighborhood sampling and aggregation, and A-DGNs adaptively adjust the graph structure. These differences affect how well the model captures the directed supply-demand relationships in the industry chain. The high performance of A-DGNs suggests that its architecture is particularly well-suited for financial datasets with heterogeneous and time-varying dependencies, as seen in A-share and CSI 300.

6.3. Ablation study

We conduct ablation experiments to explore the impact of adding industry chain information and using graph modularity regularization on model performance as shown in Table 4. There are three different settings of the ablation study: 1) “w/o IC” of Table 4 is the ablation study of the removing of industry chain information from our method, which uses linear layers to construct $\mathbf{H}$ only from the stock factor data.

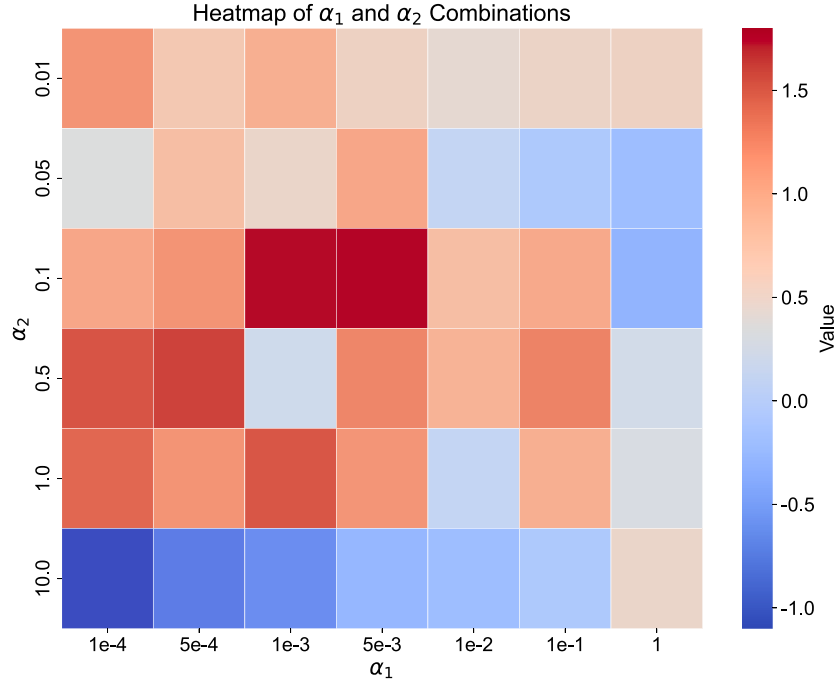


Fig. 2. Hyperparameter searching.

Table 4

Ablation Study. The first column presents three ablation settings (w/o IC: without industry chain; w/o GMR: removal of the graph modularity regularization term in Eq. (16); w/o IC and GMR: without both) alongside our proposed method.

	Sharp Ratio	Returns	Volatility	MD
w/o IC	-0.83	-0.0003	0.0183	0.3946
w/o GMR	-0.29	-0.0085	0.1193	0.7095
w/o IC and GMR	-2.467	-0.0137	0.0769	0.1291
Ours	1.56	0.0066	0.0472	0.1122

2) “w/o GMR” removes the Graph Modularity Regularization for the precision matrix $\hat{\Sigma}^{-1}$. 3) “w/o IC and GMR” represents the removal of both industry chain information and the graph modularity regularization.

The evaluation metrics include the annualized Sharpe ratio, return, and standard deviation of the annualized portfolio return, and the annualized maximum draw-down (MD).

As shown in Table 4, our method outperforms the other three ablation settings across all four evaluation metrics, particularly with a significant improvement in the annualized SR. The higher volatility in our method compared to “w/o industry chain” indicates higher portfolio risk, but it also brings greater excess returns. Specifically, removing the industry chain information results in the annualized SR dropping to -0.83, with poor values for returns and MD, indicating that incorporating industry chain relationship effectively improves the accuracy of prediction. “w/o GMR” leads to decreases in the annualized SR and returns, with volatility and MD showing the highest values across the four experiments. And “w/o industry chain and GMR” performs the worst overall, implying that industry chain relationship information significantly enhances the accuracy of the covariance matrix estimation and that the sparse covariance matrix processed with orthogonal regularization yields optimal results.

6.4. Hyperparameter selection

We conducted a coarse joint sensitivity analysis over a 6×7 grid of α_2 and α_1 values (ranging from $\{1e-4, 5e-4, 1e-3, 5e-3, 1e-2, 1e-1, 1\}$ for α_1 and $\{1e-2, 5e-2, 1e-1, 5e-1, 1, 10\}$ for α_2), resulting in

a heatmap of SR as shown in Fig. 2. This analysis reveals the joint impact, showing a peak SR at $\alpha_1 = 5e-3$, $\alpha_2 = 0.1$, suggesting a balanced regularization is optimal.

The observed sensitivity and the need for joint tuning reflect the complexity of regularizing high-dimensional covariance and return estimates. This finding highlights the importance of developing scalable optimization techniques for graph-based portfolio methods, a direction we aim to pursue in future research.

6.5. Cumulative return analysis

Cumulative return is a key metric used to measure the total return on an investment or a portfolio over a specified period. In this study, we apply the proposed method to the two stock datasets, as illustrated in Fig. 3. For the A-share dataset, we plot the cumulative return curves for four models spanning the period from November 2022 to August 2023 (see Fig. 3 (a)). The results indicate that our model performs well, even under the severe impact of COVID-19 on the global economy. Additionally, we test our model on the CSI 300 dataset to evaluate its performance prior to the COVID-19 pandemic, specifically from January 2018 to December 2019 (as shown in Fig. 3 (b)). During this period, our model also demonstrates superior performance compared to the other models.

6.6. Portfolio analysis

To emulate the commonly used portfolio sort approach, we utilize Eq. (7) for investment analysis. The first term of Eq. (7), $\text{softmax}(\mathbf{w}_t^+)$, corresponds to the long position weights of assets (labeled “High”), while the second term, $\text{softmax}(\mathbf{w}_t^-)$, pertains to the symmetric short position (labeled “Low”). Specifically, the “High” strategy retains only the first term, $\text{softmax}(\mathbf{w}_t^+)$, whereas the “Low” strategy uses the second term, $\text{softmax}(\mathbf{w}_t^-)$. The proposed method that employs both terms in Eq. (7) is labeled “High-Low”. Following the implementation details in (Feng et al., 2023), we set $a_1 = 50$ and $a_2 = 8$ in Eq. (7).

As presented in Table 5, we compute the annualized returns, annualized volatility, and annualized Sharpe ratios for four models across three experimental settings. Based on the results of three evaluation metrics,

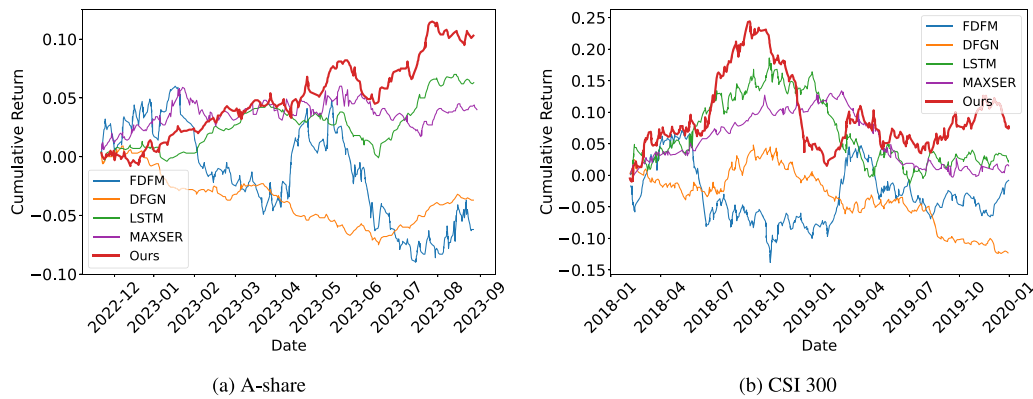


Fig. 3. The cumulative return curves on the A-Share and CSI 300 datasets.

Table 5
Portfolio analysis.

	Returns			Volatility			Sharp Ratio		
	Low	High	High-Low	Low	High	High-Low	Low	High	High-Low
FDFM	-0.0007	-0.0007	-0.0037	0.1304	0.1304	0.1303	0.2619	0.2619	0.1349
DFGN	-0.0333	-0.0181	-0.0031	0.1564	0.1305	0.0368	-3.6495	-1.9335	-1.0652
LSTM	-0.0272	-0.0188	0.0001	0.3125	0.1646	0.0018	-1.5969	-1.9446	0.9302
Ours	-0.0096	-0.0004	0.0066	0.1474	0.1557	0.0472	-0.7937	0.1825	1.5604

the performance of the “High-Low” strategy is generally superior compared to both the strategy “Low” and “High”. Specifically, it exhibits higher annualized returns, SR, and lower annualized volatility. Comparing model performance under the “High-Low” strategy reveals that the proposed model demonstrates the highest annualized returns and SR. However, its volatility falls short of LSTM and DFGN. This discrepancy may stem from the proposed model training results indicating a willingness to tolerate greater risk for higher returns.

7. Conclusion

In accordance with modern portfolio theory, the tangency portfolio weights are solely determined by the expected returns and the covariance of assets’ returns. However, estimating both of them presents significant challenges, especially in scenarios involving a large number of assets. In this work, we advocate for the integration of industry chain relationships to enhance the accuracy of covariance matrix estimation using GNNs. Despite the potential benefits, our approach has some limitations. The supply-demand relationships are inferred from the industrial chain, which only approximately represents the ground truth. Furthermore, the proposed model is difficult to tune. Financial data reflect the regulations of human economic activities, making the modeling of stock returns equivalent to modeling social regulations. This complexity likely contributes to the difficulty in tuning the proposed financial model.

CRedit authorship contribution statement

Bin Liu: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization; **Haolong Li:** Software; **Linshuang Kang:** Writing – review & editing, Supervision, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

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